



### Wonder Woman highest grossing

According to Forbes, the film's international box office has hit \$821.74 million, making it the highest grossing superhero origin film ever. <http://digit.in/WWBoxOfc>



### Hamill says no Dark Side for Luke

Luke is just "very spooked by some bad choices that led to some catastrophic consequences" <http://digit.in/LukeDark>

# Careers in Robot Cars?

Self-driving technology is cruising into India, but are we onboard?

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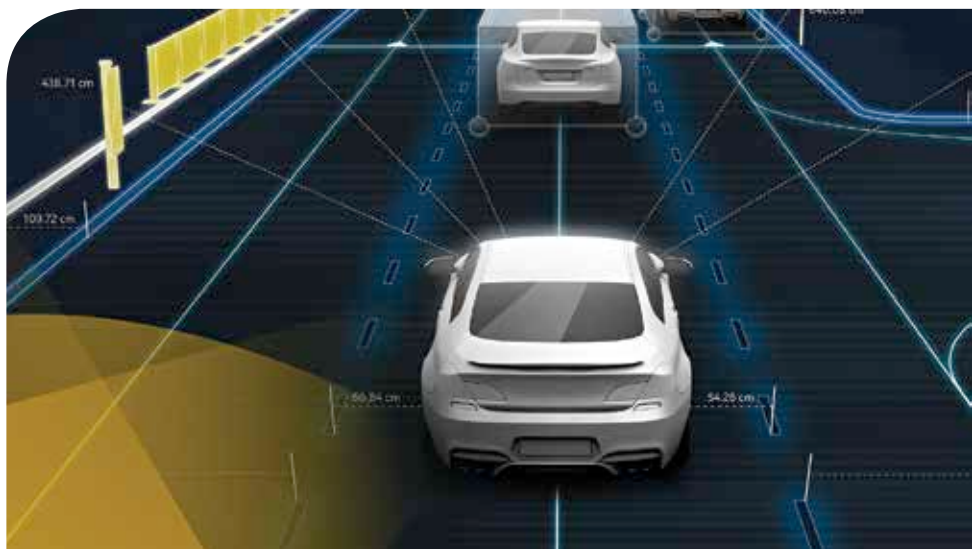
elf-driving vehicles have been a topic of contention in the world of technology for quite some time

now. While the general sentiment in the automotive industry is that autonomous cars are ready to roll into our driveways any moment now, in all their driverless glory, if reports are to be believed then truly autonomous vehicles are quite far away. Given that most people expect the industry to take a few decades to mature, acting now could get you in at the ground floor of an industry that's sure to skyrocket. After all, people like Elon Musk know a lot more about it than us. But where do you start?

## HOW IT COMES TOGETHER

Autonomous vehicles are fairly complicated systems, especially from an academic point of view, with a multitude of areas working together. Even in the industry, the main focus is on developing a problem statement that encompasses all these areas effectively – as in, the biggest question is, how to make all these systems work together effectively and seamlessly to make a flawless autonomous driving system.

For instance, the broadest field that covers most of the areas involved in autonomous driving is mobile robotics. However, there's also a huge role for mathematics. According to Ankur Pandey, Sr. Data Scientist, HERE Solutions India Pvt. Ltd., "Deep Learning (and Machine Learning in general), Computer Vision/ Image Processing, Sensors, Robotics, IoT, Embedded systems, are the hot skill



HERE Solutions is working on an HD Live map that 'self-heals' in real time

sets for breaking into self driving car/ vehicle industry. One must (also) have excellent program chops (specifically Python, and C++)."

Anyone working with autonomous vehicles has to deal with three different modules – a module to accept sensory input, another to calculate the correct outputs, which are sent to a third module – that acts on it to provide accurate guidance to the vehicle. The sensory input module would involve dealing with mechanical, visual, and digital sensors that locate the vehicle, study the surrounding terrain and objects, and pass this information on. The processing module then leverages neural networks to process this information to decide courses of action – accelerating, braking, or turning. The third module uses these decisions as an input, and convert them into mechanical actions upon the physical body of the car to translate them into the actual driving of the car.

## SKILLS NEEDED

According to Ishan Gupta, MD-India, Udacity, the skills that the industry is looking for are:

- Python and C++
- TensorFlow
- Keras
- OpenCV
- NumPy
- Scikit-learn
- Amazon Web Services
- Anaconda
- make.

Apart from this, familiarity with neural networks, image processing, sensory calibration, object detection and virtual filters, prediction and measurement loops, open and closed- loop controllers, data calibration and fusion, as well as linear quadratic regulators are also skills that make one a better candidate.

## ROBOT OS (ROS)

A good place to start, according to Rajesh Kumar, VP, Strategic initiatives